

Lab 5

A* Search

Imagine a person standing at a place called S. The person wants to reach destination called G. There are different roads between the places, and each road has a different travel cost. The person wants to find a good route to reach G without spending too much money.

From S, there are two possible roads. One road goes from S to A and costs 2. Another road goes from S to B and costs 1. From A, there is a road to G that costs 10. From B, there is a road to G that costs 4. So there are two main ways to reach the destination.

The first possible journey is $S \rightarrow A \rightarrow G$. The cost of travelling from S to A is 2, and the cost from A to G is 10. Therefore, the total cost is $2 + 10 = 12$.

The second possible journey is $S \rightarrow B \rightarrow G$. The cost from S to B is 1, and the cost from B to G is 4. Therefore, the total cost is $1 + 4 = 5$. Clearly, the second route is cheaper.

Now imagine that the person does not know the complete cost of every journey in advance. They need some way to decide which direction looks better. For this, they use an estimated distance to the destination. The estimate for A is 2, meaning A appears to be quite close to G. The estimate for B is 3, meaning B appears to be slightly farther from G. The estimate for G is 0, because G is destination.

If the person only used these estimates, they might choose A because A looks closer. But this can be a problem. The road from A to G costs 10, which is very expensive. Therefore, simply looking at the remaining distance is not enough.

A better idea is to consider two things together. The first thing is how much money has already been spent getting to the current place. The second thing is how far the destination appears to be from that place. By considering both, the person can make a more sensible decision.

For example, when the person reaches A, they have already spent 2 to travel from S to A. A also has estimated remaining distance of 2. So the person considers both the cost already spent and the estimated distance that remains.

When the person considers B, they have spent only 1 to reach B. B has an estimated remaining distance of 3. Even though B appears slightly farther from G, the amount already spent is smaller. By considering both parts together, the route through B becomes the better choice.

This is the main idea of A* search. It tries to make a better decision by considering both the journey already travelled and the estimated journey still remaining. It does not look only at where the person is now, and it does not look only at how far the destination appears to be.

We can understand this with a simple example of travelling to a shopping mall. Imagine you are trying to reach a mall. You have already travelled some distance, and there are two roads in front of you. One road looks very close to the mall, but you have already spent a lot of money getting there. Another road looks slightly farther away, but you have already spent very little money so far. Instead of looking at only one part of the journey, you consider both the money already spent and the distance still left. This gives you a better idea of which complete journey may be better.

Another example is a person planning a trip to college. Suppose one route has already required a lot of travel and still has some distance remaining. Another route has required less travel and also has a reasonable distance remaining. A person can compare both the journey already completed and the expected journey left before deciding which route to follow.

There is also an important idea about the estimated distances used in this experiment. The estimates should not be greater than the actual cost needed to reach the destination. For example, the actual cost from A to G is 10, while the estimate for A is only 2. The estimate is smaller than the actual cost, so it is safe. From B, the actual cost to G is 4, while the estimate is 3, which is also smaller. At G, both the estimate and the actual remaining cost are 0.

For S, the actual cheapest cost to reach G is 5, its estimate is 4. Again, the estimate is smaller. Therefore, all the estimates in this example were safe because they do not claim that the destination is farther than it really needs to be.

The search eventually reached G through

$S \rightarrow B \rightarrow G$

The actual cost is: $1 + 4 = 5$

The main difference between the three methods we have seen can now be understood easily. UCS mainly cares about the cost already spent. Greedy search cares about how close the destination appears to be. A* search considers both the cost already spent and the estimated cost still remaining.

Therefore, A* search can be thought of as a person who does not make a decision by looking at only one thing. They consider where they have already travelled, how much it has cost, and how far the destination appears to be. By combining these two informations, A* search can find an efficient route and, when it estimates we are safe in the required way, can find the least-cost route.